



MAJOR PROJECT • 7TH SEMESTER

Automatic Door Operating System Using PIR Sensor

An embedded automation system built on Arduino UNO for touch-free door control

NAME

SEMESTER

DEPARTMENT

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7th Semester

ECE (B.Tech)

Project Developer

Project Overview



Designed an embedded system using a PIR sensor and Arduino UNO to automate door operation — enabling hands-free, hygienic, and energy-efficient access in public and private spaces.



Embedded Core

Arduino UNO R3 processes real-time motion signals



Mechanical Actuation

Servo motor drives smooth, controlled door movement



Energy Efficient

Operates only on motion detection, minimizing power use

Why Automate Door Operation?



The Core Challenge

Manual door operation is inconvenient, unhygienic, and inaccessible in high-traffic public spaces — creating friction, contact exposure, and barriers for people with limited mobility.



Constant manual contact

Hygiene risk in shared public spaces



Accessibility gaps

Difficult for elderly & disabled users



No automation

No smart, low-cost touch-free solution

System Components



Arduino UNO R3

Core microcontroller for signal processing



Servo Motor MG996R

High-torque actuator for door movement



PIR Sensor

Detects human motion via infrared



LED Indicator

Visual status feedback for door state



Jumper Cables

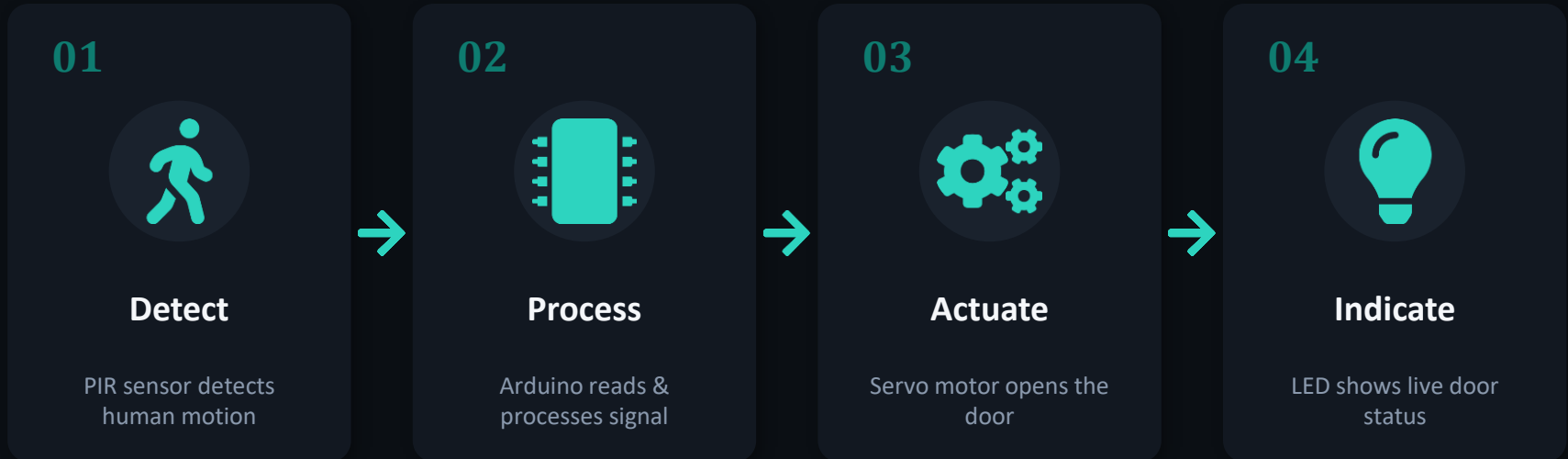
Circuit interconnections



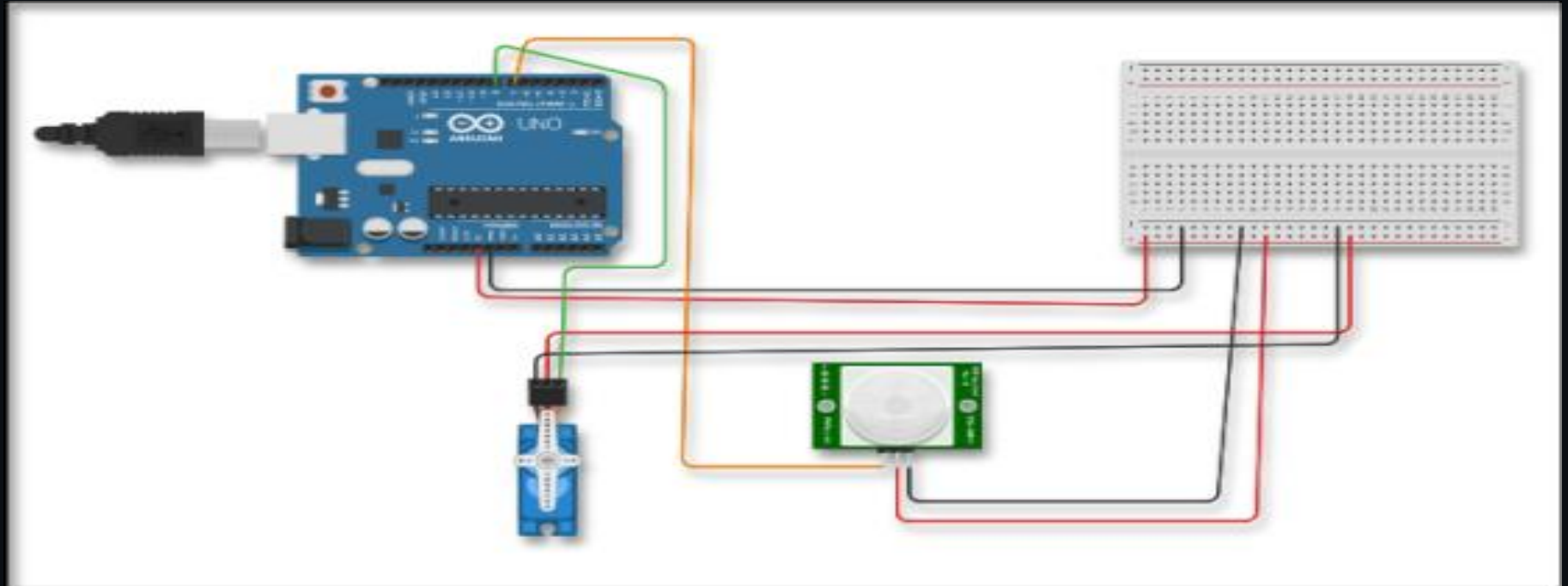
Motor Driver IC

Regulates power to the servo motor

Working Principle



Circuit Diagram



LOGIC FLOW

Flowchart



NO → Door stays closed; system keeps monitoring

YES → Door stays open; delay timer resets



Logic mirrors the project's PIR threshold flow: the sensor output is set High/Low, the door opens to the servo's preset angle on detection, and closes only after the hold delay if no further motion is sensed.

Implementation & Results

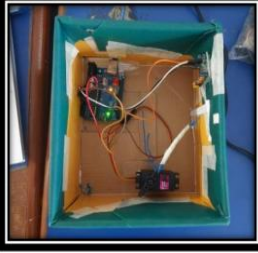


Fig.9.1

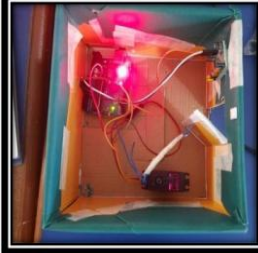


Fig. 9.2

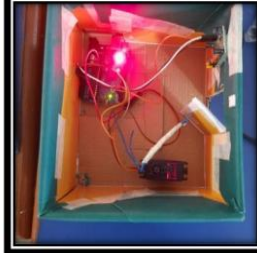


Fig. 9.3

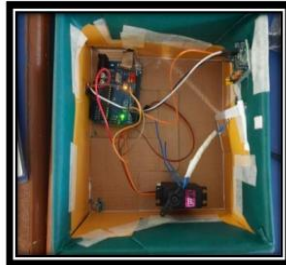


Fig. 9.4

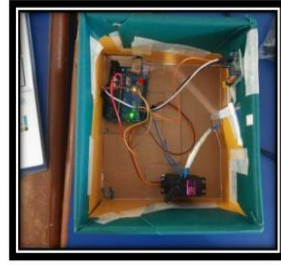


Fig. 9.5

ARDUINO SKETCH

```
if (digitalRead(PIR_PIN) == HIGH) {  
  digitalWrite(LED_PIN, HIGH);  
  servo.write(90);    // open door  
  delay(5000);        // hold open  
  servo.write(0);     // close door  
  digitalWrite(LED_PIN, LOW);  
}
```


Applications



Shopping Malls

Smooth, high-traffic entry
& exit



Hospitals

Hygienic, contact-free
access



Theatres

Convenient crowd
movement



Accessibility

Easier access for disabled
persons

WHAT'S NEXT

Future Scope



Radar Sensor Integration

More precise & longer-range motion detection



Biometric Access

Fingerprint / face recognition for secure entry



IoT-Based Control

Remote monitoring & control via cloud connectivity

CONCLUSION

Key Takeaways



Low-Cost

Built with affordable,
accessible components



Energy-Efficient

Motion-triggered operation
saves power



Reduces Effort

Removes need for manual
door handling



Scalable

Adaptable for smart
building integration

A practical step toward smart, accessible building automation.

SOURCES

References

- 1 Arduino UNO R3 — Official Documentation, docs.arduino.cc/hardware/uno-rev3
- 2 PIR Sensor Tutorial — Last Minute Engineers, lastminuteengineers.com/pir-sensor-arduino-tutorial
- 3 Servo Motors — Arduino Docs, docs.arduino.cc/learn/electronics/servo-motors
- 4 Servo Motor Tutorial — Last Minute Engineers, lastminuteengineers.com/servo-motor-arduino-tutorial
- 5 Automatic Door Opening/Closing System using PIR Sensor — CIT Central Library Project Report Archive

SKILLS DEMONSTRATED

Resume Impact



Embedded C / Arduino Programming

Wrote and debugged real-time control logic for sensor-driven automation



Sensor Integration (PIR, Servo)

Interfaced motion sensors and actuators with a microcontroller



Circuit Design & Prototyping

Designed, wired, and tested a functional hardware prototype



Real-World Automation Application

Applied embedded systems concepts to solve a practical accessibility problem